Homework 9

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ECE 3340 Numerical Methods

Homework 9: Properties of Linear Systems

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Problem 1: LUP Decomposition

Calculate the LUP decomposition of the following matrix:

$$\mathbf{A} = \begin{bmatrix} -4 & -2 & 0 \\ 6 & -6 & -7 \\ -4 & 2 & -7 \end{bmatrix}$$

$\ell_{21} = -\frac{3}{2}$

$\ell_{31} = \frac{-4}{-4} = 1$

$R_2 \leftarrow R_2 - \ell_{21} R_1 = R_2 + \frac{3}{2} R_1$

$[6,-6,-7] + \frac{3}{2}[-4,-2,0] = [6,-6,-7] + [-6,-3,0] = [0,-9,-7]$

$R_3 \leftarrow R_3 - \ell_{31} R_1 = R_3 - R_1$

$[-4,2,-7] - [-4,-2,0] = [0,4,-7]$

$\begin{bmatrix} -4 & -2 & 0 \\ 0 & -9 & -7 \\ 0 & 4 & -7 \end{bmatrix}$

$a_{22} = -9$

$\ell_{32} = \frac{4}{-9}$

$R_3 \leftarrow R_3 - \ell_{32} R_2 = R_3 + \frac{4}{9} R_2$

$[0,4,-7] + \frac{4}{9}[0,-9,-7] = [0,4,-7] + [0,-4,-\frac{28}{9}] = [0,0,-\frac{91}{9}]$

$U = \begin{bmatrix} -4 & -2 & 0 \\ 0 & -9 & -7 \\ 0 & 0 & -\frac{91}{9} \end{bmatrix}$, $L = \begin{bmatrix} 1 & 0 & 0 \\ -3/2 & 1 & 0 \\ 1 & -4/9 & 1 \end{bmatrix}$ $P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$\mathbf{L} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{3}{2} & 1 & 0 \\ 1 & -\frac{4}{9} & 1 \end{bmatrix} \qquad \mathbf{U} = \begin{bmatrix} -4 & -2 & 0 \\ 0 & -9 & -7 \\ 0 & 0 & -\frac{91}{9} \end{bmatrix} \qquad \mathbf{P} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Problem 2: Matrix Inversion

Calculate the empty column of $\mathbf{A}^{-1}$ using your previous LUP decomposition.

$A^{-1} = \begin{bmatrix} -\frac{2}{13} & \square & -1/26 \\ -5/26 & \square & 1/13 \\ 3/91 & \square & -9/91 \end{bmatrix} \Rightarrow A^{-1} = \begin{bmatrix} -2/13 & 1/26 & -1/26 \\ -5/26 & -1/13 & 1/13 \\ 3/91 & -4/91 & -9/91 \end{bmatrix}$

$$\mathbf{A}^{-1} = \begin{bmatrix} -2/13 & \boxed{1/26} & -1/26 \\ -5/26 & \boxed{-1/13} & 1/13 \\ 3/91 & \boxed{-4/91} & -9/91 \end{bmatrix}$$

$$\begin{bmatrix} 3/91 & \boxed{-4/91} & -9/91 \end{bmatrix}$$

Problem 3: Condition Numbers

The condition number of a Hilbert matrix scales as $O\left(\frac{(1+\sqrt{2})^{4n}}{\sqrt{n}}\right)$. What is the largest Hilbert matrix that can be numerically solved using IEEE octuple (256-bit) precision (1 signed bit, 236-bit mantissa, 19-bit exponent) while still providing at least a 32 bits of precision? $O\left(\frac{(1+\sqrt{2})^{4n}}{\sqrt{n}}\right)$ $236-32=204$ bits

$\kappa(H_n) \le 2^{204}$

$\kappa(H_n) \approx \frac{(1+\sqrt{2})^{4n}}{\sqrt{n}}$

$4n\log_2(1+\sqrt{2}) - \frac{1}{2}\log_2 n \lesssim 204$

$\log_2(1+\sqrt{2}) \approx 1.27155$

$5.0862n - \frac{1}{2}\log_2 n \lesssim 204$

$n=40$: $5.0862(40) - \frac{1}{2}\log_2(40) \approx 200.8 < 204$

$n=41$: $5.0862(41) - \frac{1}{2}\log_2(41) \approx 205.9 > 204$

$\boxed{n = 40}$